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Surface and interface analysis for copper phthalocyanine (CuPc) and indium-tin-oxide (ITO) using X-ray photoelectron spectroscopy (XPS)

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Abstract

Indium-tin-oxide (ITO) has been widely used as a hole injection electrode for organic light-emitting devices (OLEDs), but the work function of ITO film usually mismatches the highest occupied molecular orbital (HOMO) of the hole transport materials. Copper phthalocyanine (CuPc) has been used as a hole injection buffer to enhance the hole injection from ITO to hole transport layer (HTL). A thin CuPc layer was thermally evaporated onto ITO-coated glass substrate and the surface and interface electron states of the CuPc/ITO close contact were measured and analyzed by X-ray photoelectron spectroscopy (XPS) technology. Results show that, in the interface of CuPc and ITO film, the role of oxygen is very crucial, the oxygen in ITO film may diffuse into the organic semiconductor and cause the alteration of electron states in organic materials.

Keywords: OLEDs; Surface and interface analysis; X-ray photoelectron spectroscopy (XPS)

1. Introduction

Since the first multi-layered organic light-emitting devices (OLEDs) were reported [1], OLEDs have attracted extensive attention due to their excellent performance with high brightness, low power consumption, wide viewing angle, fast response time and potentially low cost [1,2]. Multilayered OLEDs usually consist of anode, hole transport layer (HTL), light-emitting layer, electron transport layer (ETL) and cathode. Indium-tin-oxide (ITO) coated glass has been widely used as an anode (i.e. hole injection electrode) for OLEDs because ITO thin film has high transparency and low resistivity [3]. But the work function of ITO is only 4–5 eV (typically 4.5–4.7 eV) and depends on the surface treatments [4]. For most organic semiconductor materials, the highest occupied molecular orbitals (HOMOs) lie more than 5 eV below vacuum. For example, *N*,*N*′-bis-(1-naphthyl)-*N*,*N*′-diphenyl-4,4′-diamine (NPB) and *N*,*N*′-diphenyl-*N*,*N*'-bis-(3-methylphenyl)-1,1′-biphenyl-4,4′-diamine (TPD), which are widely used as hole transport materials, have HOMO of 5.2 and 5.5 eV below vacuum, respectively [5,6]. It is well known that, the electroluminescence efficiency of OLEDs is related to their interface characteristics. The mismatch

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between the work function of ITO and the ionization potential of HTL may result in an injection barrier between the ITO anode and the HTL is formed during operation. In order to inject holes effectively, the barrier between ITO anode and HTL cannot be too large. Some organic materials have been adopted as hole injection buffer layers inserted between the ITO anode and HTL to enhance the performance of OLEDs, and copper phthalocyanine (CuPc) is the common hole injection buffer material [7–16]. The buffer layer can improve the performance of OLEDs from several aspects, such as suppressing noisy leakage current, reducing the operating voltage, and enhancing the thermal stability and quantum efficiency [17].

Copper phthalocyanine has been widely used in OLEDs as hole injection buffer layer because of the good match of its HOMO (5.1 eV below vacuum [18]) to the work function of ITO. Concurrently, the poor solubility of CuPc in common organic solvents enables the deposition of polymer films onto it by conventional spin casting. Moreover, CuPc has very weak absorption of light with wavelengths from 400 to 500 nm, making it suitable for use in blue and green OLEDs [12]. In more recent years, many OLEDs with CuPc have been studied. Scott and co-workers [19] reported that the oxygen released from ITO film would degrade the hole transport materials, which results in the formation of luminescence quenching centers, and increases the operational voltage of the OLEDs. So CuPc was inserted between the ITO anode and HTL, could reduce the ITO-induced degradation on the hole transport materials. Moreover, Nuesch et al. [20] investigated the role of CuPc for the injection of holes from ITO anode into an HTL. They found that, once a thin CuPc layer is used at the ITO interface, the oxygen in ITO diffuses into CuPc and an interfacial space charge layer is formed, which results in the effective ITO work function is pinned at the HOMO orbital energy of CuPc and the injection efficiency of charge carriers does not depend significantly on the surface treatment of the ITO electrode [20].

To further understand the ITO-induced degradation and the role of CuPc in OLEDs, we investigate the surface and interface electron states of CuPc/ITO contact using X-ray photoelectron spectroscopy (XPS) technology.

2. Experiment

The CuPc/ITO sample was fabricated with traditional vacuum deposition method. The substrate used in the present experiment was ITO-coated glass. The sheet resistance and thickness of the ITO film were about 40 Ω/□ and 50 nm, respectively. Prior to the organic deposition, the ITO-coated glass substrate was routinely cleaned by ultrasonic treatment in detergent solution, followed by through rinsing in de-ionized water, and then boiled in acetone followed by methanol. After being dried in a vacuum oven, the substrate was immediately transferred into a deposition chamber. The distance from the substrate to the evaporative source was 15 cm. The CuPc was thermally evaporated onto the ITO-coated glass substrate at a vacuum about 5×10^{-5} Pa and room temperature. The thickness of CuPc thin film was about 20 nm.

The surface and interface electron states of the CuPc/ITO contact were measured and analyzed with an ESCALAB220-IXL type X-ray photoelectron spectroscopy instrument with a base pressure of better than 1.0×10^{-8} Pa (made in VG company in England). Mg Kα radiation ($h\nu = 1253.6$ eV) was used as X-ray source and operated at 300 W. The pass energy of the spectrometer with a resolution of 0.74 eV was 30 eV and the scanning step was 100 meV. In order to obtain its interface characteristics, the CuPc/ITO sample was sputtered by argon ion beam with a kinetic energy for Ar^+ of 3.0 keV, at 3×10^{-7} Pa, Ar pressure in a vacuum chamber as described above. In general, argon ion sputtering may damage the surface of the CuPc/ITO sample, and may result in shifts of core-level and broadening of spectral peaks. But we can diminish the sputter effect through large area and low energy argon ion sputtering [29]. The large sputtering area of $0.75 \times 0.75\ cm^2$ was adopted to decrease the damage

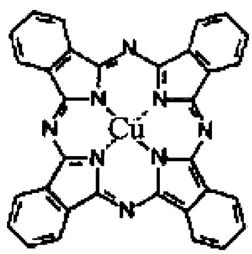

Fig. 1. The chemical structure of copper phthalocyanine (CuPc) molecule.

of argon ion sputtering effect to the CuPc structure. The ion beam density was about 1.0 μA/(0.75 × 0.75) cm^2.

3. Results and discussion

Figs. 1 and 2a shows the XPS whole scanning spectrum recorded at the raw surface of the CuPc/ITO contact. There were six relatively strong peaks in the XPS whole scanning spectrum. The peaks located at about 937, 402 and 290 eV correspond to the electron states of Cu 2p, N 1s and C 1s in the CuPc molecule, respectively. The O 1s peak is approximately located at 534 eV. The two peaks close to 450 eV correspond to the electron states of In $3d_{3/2}$ and In $3d_{5/2}$. The Sn 3d peak was relatively weak. The

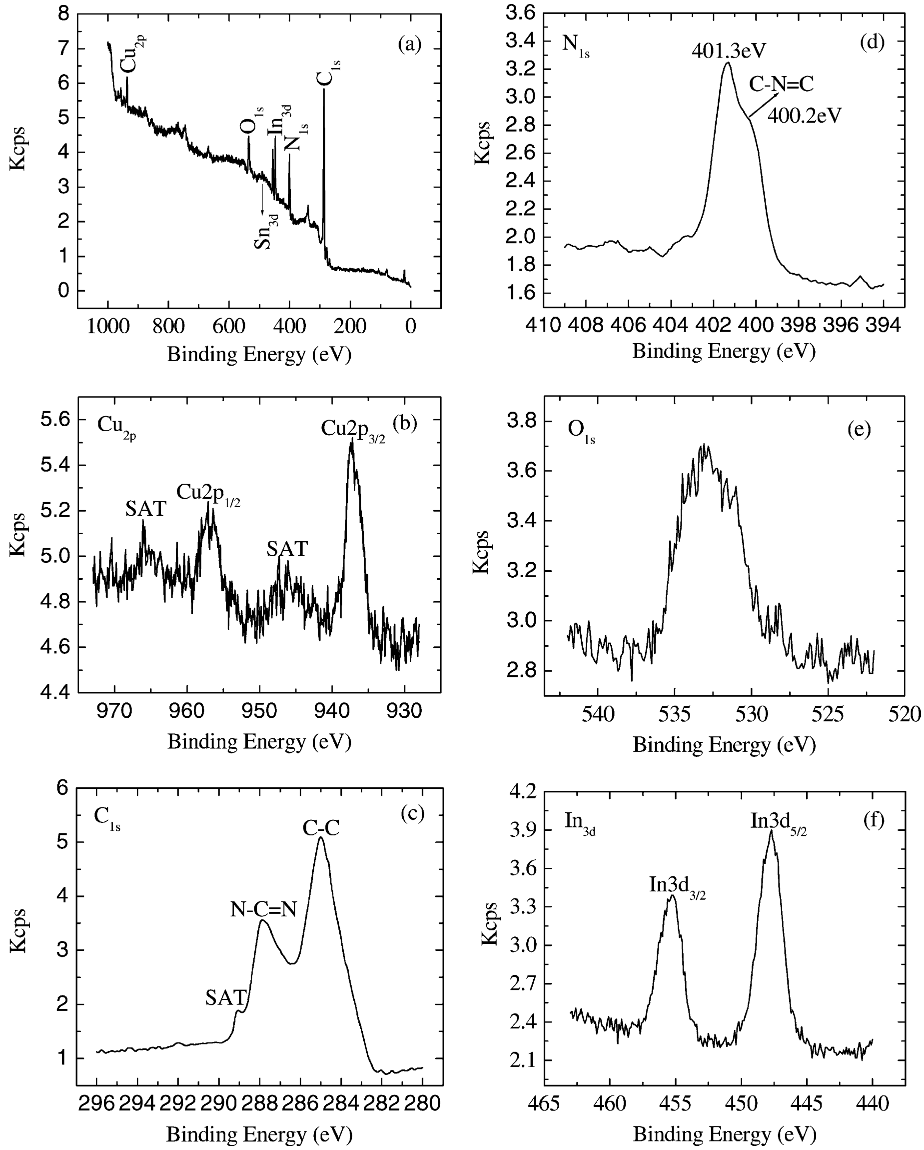

Fig. 2. The XPS whole scanning spectrum (a) and the fine spectra of Cu 2p (b), C 1s (c), N 1s (d), O 1s (e) and In 3d (f) recorded at the raw surface of the CuPc/ITO contact. (SAT—satellite peak).

above results show that the CuPc film was very thin, so the secondary electrons of oxygen and indium atoms in the ITO film were excited when the CuPc film surface was scanning. But the content of tin atoms was much less than that of indium atoms, this results in the peak of Sn 3d was much weaker than that of In 3d. From Fig. 2a we can find that the Sn:In atomic ratio near the surface of ITO film is about 1:9, this value is close to the one obtained by Wu et al. [21]. Besides these peaks, there were some weak peaks obtained from the glass substrate under the ITO thin film.

To further investigate into the electron states of the CuPc and the underlying ITO thin film, the XPS fine spectra of Cu 2p, C 1s, N 1s, O 1s and In 3d are given in Fig. 2b–f, respectively. From Fig. 2b we can see that the Cu 2p spectrum had two strong peaks, one located at 957.1 eV and another located at 937.2 eV, corresponding to the electron states of Cu $2p_{1/2}$ and Cu $2p_{3/2}$, respectively [16]. The binding energy of copper atoms (in Cu $2p_{3/2}$ electron state) was 937.2 eV, showing that the copper atoms in CuPc thin film were in Cu(II) [22]. This can be explained from the chemical structure of CuPc molecule (Fig. 1). In the CuPc molecule, the copper atom bonded with nitrogen atoms through coordinate bonds. Moreover, the Cu(II) species with multiplet splitting satellite were observed, which reflect the $3dx^2 - y^2$ orbital contained in the molecular plane. From the fine spectrum of C 1s in the raw surface state (Fig. 2c) we can find that, the main peak located at 285.0 eV, which corresponds to the binding energy of the aromatic carbon atoms, i.e. the carbon atoms bonding with carbon and hydrogen atoms [23,24]. The binding energy of carbon atoms in C–N bonds was observed at 287.9 eV [22]. We know that the exact binding energy of the electron of certain elements relates to the chemical environment of the elements. According to the chemical environment there are two kinds of carbon atoms: 8 carbon atoms bonding with 2 nitrogen atoms and the other 24 carbon atoms of an aromatic hydrocarbon character, and the number ratio of the carbon atoms of an aromatic hydrocarbon character to the carbon atoms bonding with nitrogen atoms is 3:1. The binding energy of the carbon atom is affected by the electronegativity of the atom bonding with it. The larger the electronegativity of the related atom, the smaller the electron density around the carbon atom, and the binding energy of the carbon atom gets bigger as the electronegativity of the related atom increases. The electronegativity of carbon and hydrogen is less than that of nitrogen, so the binding energy of the carbon atoms of an aromatic hydrocarbon character is less than that of the carbon atoms bonding with nitrogen atoms. We can still find that the area ratio of C–C peak to C–N peak was nearly 3:1, which is in agreement with the atomic ratio of the two kinds of carbon atoms. Besides these, a satellite peak was observed at 289.1 eV, this binding energy is slightly more than that of carbon atom bonding to one carbon and two nitrogen atoms [25], which was thought to be due to oxidation, and the oxygen atoms may be originated from the underlying ITO film and the O_2 absorbed by the CuPc film. For the N 1s spectrum (Fig. 2d), the main peak located at 401.3 eV, and a shoulder peak emerged at 400.2 eV. In the CuPc molecule, the nitrogen atoms were also in two kinds of chemical environment: four nitrogen atoms only bond with two carbon atoms and form C–N=C bonds, and the other four nitrogen atoms not only bond with carbon atoms but also bond with copper atom through coordination bond. We consider that the peak located at 400.2 eV corresponds to the binding energy of the nitrogen atoms in C–N=C bonds [26], and the peak observed at 401.3 eV may be due to the nitrogen atoms which bonding with carbon and copper atoms. The O 1s spectrum was not very regular (Fig. 2e), and the peak of O 1s spectrum located at a very broad scale of 530–536 eV. The oxygen atoms mainly originated from ITO film, but the binding energy of oxygen atoms in In_2O_3 and SnO_2 is about 530.3–530.6 eV [27,28]. So we consider that in the process of being transferred from the vacuum deposition chamber into the XPS analyzer, the sample was exposed in the atmosphere, some O_2 and H_2O adsorbed by the sample diffuse into the CuPc and ITO film, which results in the O 1s spectrum peak broadens towards higher binding energy, because the oxygen in O_2 and H_2O has higher binding energy [22]. Concurrently, the oxygen in ITO film might diffuse into CuPc and the interaction of oxygen with copper, carbon and nitrogen would result in the broadening of O 1s spectrum peak. For the In 3d spectrum (Fig. 2f), there were two splitting energy scale: In $3d_{3/2}$ peak located at 455.2 eV and In $3d_{5/2}$ peak was observed at about 448.0 eV. The binding energy corresponding to the In $3d_{5/2}$ was higher than that of indium atoms in In_2O_3 [27], which may be

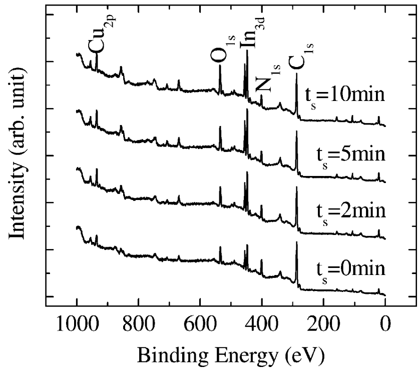

Fig. 3. The evolution of the XPS whole scanning spectra for the CuPc/ITO contact as the sputtering time increases.

caused by the oxygen diffused from atmosphere into ITO.

In order to investigate interface electron states, the CuPc/ITO sample was sputtered by argon ion beam. The XPS spectra data acquisition for the CuPc/ITO sample was carried out when the sputtering time reached 2, 5 and 10 min, respectively. The evolution of the XPS whole scanning spectra for the CuPc/ITO as the sputtering time increases is shown in Fig. 3. The XPS whole scanning spectrum of the raw surface is also given in the same figure to compare it with that of the interface. We can find that, as the sputtering time increases, the peaks of C 1s and N 1s gradually become weaker, but those of In 3d and O 1s slowly get strong. We know that, the CuPc film on the ITO-coated glass substrate becomes thinner and thinner with the increasing sputtering time, the number of carbon and nitrogen atoms gradually reduced, so less and less secondary electrons of carbon and nitrogen atoms were excited, which results in the peaks of C 1s and N 1s gradually become weaker. But the excitation of more and more indium and oxygen in underlying ITO film makes the intensity of In 3d and O 1s peaks increase. Theoretically, as the interface of CuPc and ITO films is approached, the intensity of Cu 2p peak should reduce because of the decrement of copper atomic concentration, but in fact, the Cu 2p peak almost stays the same. This may originate from the copper atoms detached themselves from the chemical bonds in CuPc molecules diffused into the interface of CuPc/ITO. When the CuPc/ITO sample was bombarded by argon plasma, the increment of its surface temperature may results in

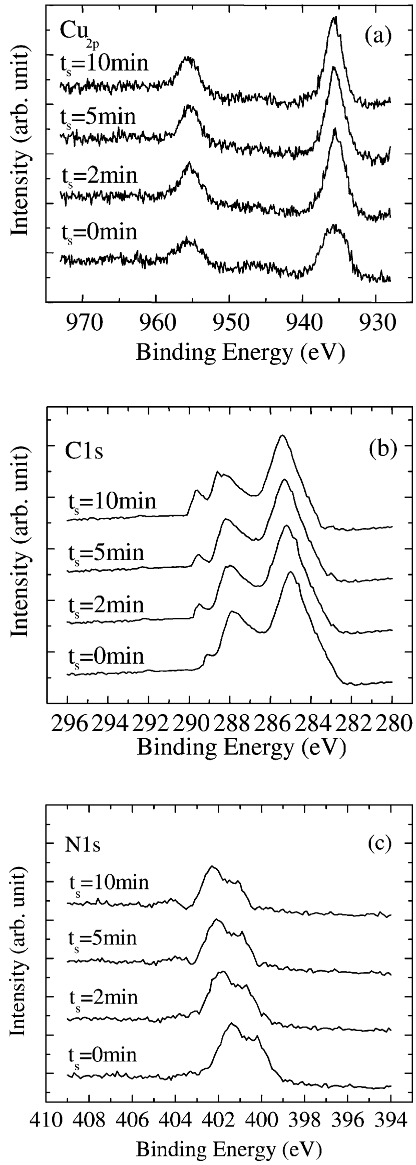

Fig. 4. The evolution of the XPS fine spectra of Cu 2p (a), C 1s (b) and N 1s (c) for the CuPc/ITO contact as the sputtering time increases.

the copper atoms bonding to nitrogen atom through coordination bond detach from Cu–N bonds and diffuse into the interface of CuPc/ITO [25]. However, the interaction of copper with oxygen atoms makes the Cu 2p peak almost does not shift.

To further understand the interface electron state of the CuPc/ITO, the evolution of the XPS fine spectra of

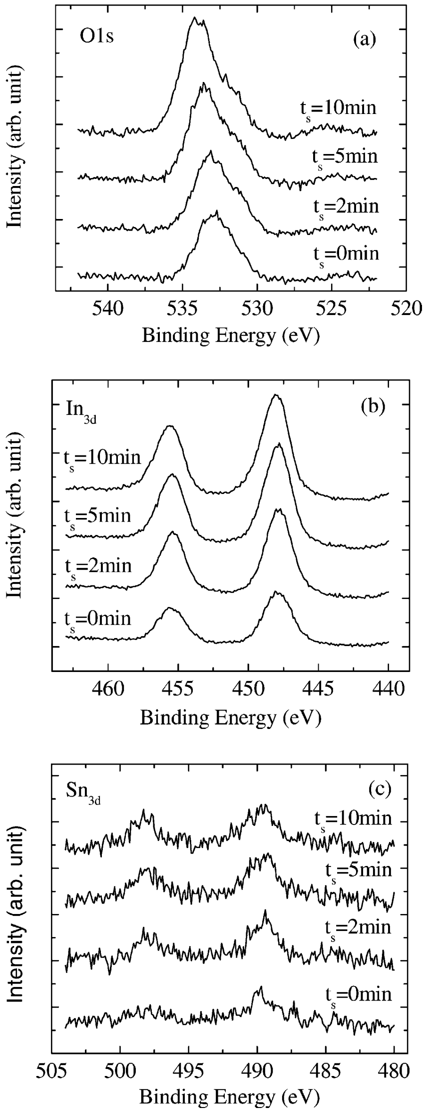

Fig. 5. The evolution of the XPS fine spectra of O 1s (a), In 3d (b) and Sn 3d (c) for the CuPc/ITO contact as the sputtering time increases.

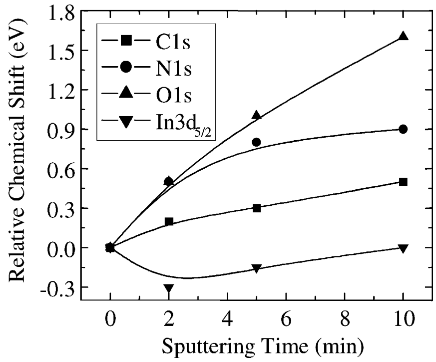

Fig. 6. The relative chemical shifts of C 1s, N 1s, O 1s and In $3d_{5/2}$ for the CuPc/ITO contact as the sputtering time increases.

the Cu 2p, C 1s, N 1s, O 1s, In 3d and Sn 3d are shown in Figs. 4 and 5, respectively. Fig. 4 shows that, when the sputtering time increases, the intensity of Cu 2p peaks almost keeps constant and the peak position also has no obvious alteration. For the C 1s and N 1s spectra, the peak area gradually reduces and the peak positions slowly shift towards higher binding energy. Fig. 6 shows the relative chemical shifts of C 1s, N 1s, O 1s and In $3d_{5/2}$ as a function of argon plasma bombardment time. The alteration of peak positions of C 1s and N 1s may be due to the effect of oxygen originated from the ITO film and absorbed O_2 and H_2O. As the interface of CuPc/ITO is approached, the concentration of oxygen mainly originating from the ITO film slowly increases, and the interaction of oxygen with carbon and nitrogen atoms strengthens, which causes the C 1s and N 1s peaks shift towards higher binding energy. Moreover, from Fig. 4b we can find that the satellite peak at 289.1 eV is growing with increasing sputtering time, which further indicates that more oxygen is present close to the ITO interface. The core-levels of O 1s spectra slowly shift towards higher binding energy, which similar to that of C 1s and N 1s spectra, but the peak area of O 1s spectra gradually increases (Figs. 5a and 6c). For the In 3d and Sn 3d spectra (Fig. 5b and c), the peak area gradually increases with increasing sputtering time, but the core-levels firstly shift towards low binding energy then shift back to high binding energy. This may be also due to the effect of oxygen. When the sputtering time increases, the oxygen maintained in the ITO film slightly reduces because of the selective sputtering effect of argon ions [29], which results in the reductive effect of In_2O_3 and SnO_2. Because the suboxides of indium and tin have relatively lower binding energy, the core-levels of In 3d and Sn 3d spectra shift towards low binding energy. When the sputtering time continuously increases, the concentration of oxygen then increases and tends to balance, and the suboxides of indium and tin reduce, which results in the core-levels of In 3d and Sn 3d spectra shift back to high binding energy. Because the behavior of Sn 3d spectra is similar to that of In 3d spectra, Fig. 6 does not give the relative chemical shifts of Sn 3d. From Fig. 6 we can also find that, as the sputtering time increases, the chemical shifts of the O 1s is most obvious, this also verifies the effect of the selective sputtering effect of argon ions on oxygen.

4. Conclusion

In conclusion, we have investigated the surface and interface electron states of the CuPc/ITO close contact using XPS technique. The analysis indicates that in the interface of CuPc and ITO film, the role of oxygen is very crucial. When the ITO film contacts with organic materials, the oxygen in ITO film may diffuse into the organic materials and result in the degradation of organic and even OLEDs.

Acknowledgements

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